

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: EU Pharmaceutical Regulatory NLP — French

Definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Justification

DrBenchmark's task taxonomy (POS, NER, MCQA, MCC, MLC, STS) [Q81] is calibrated for clinical and biomedical research NLP rather than the EU regulatory workflow. The deployment requires NER and STS over EMA/ANSM regulatory document genres (SmPCs, CTD modules, EU-RMPs, PILs), and these document categories do not appear as explicit task categories in the benchmark. While QUAERO EMEA and Mantra-GSC EMEA/Patents provide partial PIL/patent overlap [Q34, Q40], the majority of tasks (FrenchMedMCQA, MORFITT, DiaMED, CAS POS, ESSAI POS, E3C) involve task types or genres orthogonal to the deployment's NER/STS regulatory compliance functions. This is a HIGH-priority dimension per the elicitation, and the gap is substantial — though not total, since QUAERO and CLISTER provide partial NER and STS coverage on biomedical French.

Strengths

- Task inventory does include both NER and STS — the two task types the deployment needs [Q1, Q4, Q18]
- QUAERO EMEA NER provides authentic EMA package leaflet text in regulatory register, the only confirmed regulatory-genre source (QUAERO-D1, QUAERO-D2, QUAERO-D14)
- Mantra-GSC includes EMEA and Patents NER configurations that are documented as deployment-relevant genres [Q40]
- The benchmark's diversity across 20 tasks gives broad signal for French biomedical model selection [Q83]

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: The deployment requires NER (entity extraction over INNs, ATC codes, excipients, posology, contraindication qualifiers) and STS (regulatory equivalence scoring for safety warnings) over SmPCs, PILs, CTD modules, and EU-RMPs. These task categories partially exist in the benchmark (NER, STS) but are calibrated for clinical/research text, not regulatory genres [Q81], [WEB-7].

ID	Evidence
WEB-7	EMA confirms SmPC/PIL templates as the regulatory document standard

IO-2: Regulatory document genres (SmPC, CTD, EU-RMP) are absent as explicit task categories. The benchmark's only confirmed regulatory-genre source is QUAERO EMEA (~10 segmented PIL documents) [WEB-3, WEB-4]. SmPC, CTD module, and EU-RMP text are not documented as data sources [Q34]. Regulatory equivalence as a distinct STS sub-task is absent.

ID	Evidence
WEB-3	QUAERO HuggingFace confirms EMEA = patient leaflet text

WEB-4	QUAERO official corpus page documents 103,056 words across drug leaflets and biomedical titles
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IO-3: Several included tasks are largely orthogonal to the deployment: FrenchMedMCQA (closed-option exam QA) (FRENCHMEDMCQA-D11, FRENCHMEDMCQA-D13), MORFITT (specialty classification of research abstracts) (MORFITT-D6, MORFITT-D8), DiaMED (ICD-10 chapter classification of African clinical cases) (DIAMED-D1, DIAMED-D9), and CAS/ESSAI POS-only configs (CAS-D13, ESSAI-D18). These contribute to benchmark scores but consume evaluation effort without testing deployment-relevant capabilities.

ID	Evidence
FRENCHMED MCQA-D11	Au cours de la leucémie lymphoïde chronique, le myélogramme montre: Une population de lymphocytes>30% — Closed-option recall task; no NER span or STS output produced
MORFITT-D6	La région du Nord-Ouest de l'Ontario présente un taux élevé...d'infections de la peau et des tissus mous causées par une souche de Staphylococcus aureus méthycillino-résistante d'origine communautaire — Research abstract with no entity-level or similarity ground truth; supports task-type mismatch concern
DIAMED-D1	Le traitement était basé sur les antirétroviraux (Stavudine (D4T), Lamuvidine (3TC), Efavirenz (EFV)) et une chimiothérapie (Doxorubicine 50 mg/j en cure tous les 19 jours puis Tamoxifène) — Drug names with dosage in clinical narrative; informal INN/abbreviation hybrid, not EMA-standardized SmPC format
CAS-D13	un traitement empirique de méropénème et de doxycycline est introduit — Drug INNs present but annotated only for POS, not regulatory entities
ESSAI-D18	Son avantage est de réduire les effets indésirables tout en continuant la même dose intensité . — "effets indésirables" and "dose intensité" present but only POS-tagged, not entity-labelled

IO-4: Documented gaps: no SmPC genre, no CTD module narrative tasks, no EU-RMP tasks, no IDMP-aligned structured-extraction tasks [WEB-16, WEB-17], no regulatory-equivalence STS sub-task. These omit core categories needed for content validity in the EU pharmaceutical regulatory deployment.

ID	Evidence
WEB-16	IQVIA NLP regulatory affairs factsheet confirms IDMP structured-extraction requirements
WEB-17	Linguamatics IDMP blog confirms ~70% of IDMP attributes reside in unstructured text

Information Gaps

- Whether any French NLP benchmark targeting SmPC/CTD/EU-RMP genres exists outside DrBenchmark — searches found none (gap_id 7 RESOLVED as confirmed gap)
- Pair count and source-genre split for DEFT-2020 STS subtask not stated in the paper

Requires Expert Verification

- Whether the deployment's CTD module supplementary template layer can fully compensate for the absence of CTD-genre evaluation in the benchmark

Remediation

Gap 1: SmPC, CTD module, and EU-RMP document genres are absent as task categories; only QUAERO EMEA (~10 segmented PIL documents) provides confirmed regulatory-genre coverage.

Recommendation: Use QUAERO EMEA only as a partial sanity check for PIL-register NER. For SmPC/CTD/EU-RMP genres, construct a deployment-internal evaluation corpus from authentic regulatory submissions under current QRD v10.4 [WEB-7] and treat DrBenchmark performance as a model-selection proxy, not a deployment-readiness measure.